JOURNAL OF ANIMAL SCIENCE
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J ANIM SCI 1992, 70:2000-2012.

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Estimation of Effects of Single Genes on Quantitative Traits[1]

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ABSTRACT: Studies involving the effects of single genes on quantitative traits may involve closed populations, selection may be practiced, and the quantitative trait of concern may also be influenced by background genes that are inherited in a polygenic manner. It is shown analytically that analysis of such data by ordinary least squares, the usual method of analysis, can lead to finding an excess of spurious significant effects of single genes, when no effect exists, for both randomly and directionally selected populations and can lead to bias in estimates of single-gene effects when selection has been practiced. The bias depends on heritability of the polygenic effects on the trait, selection intensity, mode of inheritance, magnitude of gene effect, gene frequency, and data structure. It is argued that when genotypes of individuals can be identified for all individuals with observations on the trait, use of mixed-model procedures under an animal model treating single-gene effects as fixed effects can provide unbiased estimates of single-gene effects and exact tests of associated hypotheses for pedigreed populations, even when selection is practiced. Results are illustrated through computer simulation.

Key Words: Major Genes, Least Squares, Best Linear Unbiased Prediction, Analysis of Variance

J. Anim. Sci. 1992. 70:2000–2012

Introduction

A number of single genes, inherited in a simple Mendelian manner, that can be screened for and identified in the animal have been shown to be associated with quantitative traits of economic importance. Examples are associations of genetic variants of casein with milk production traits (Ng-Kwai-Hang et al., 1984), halothane reaction with meat yield and quality in pigs (Jensen and Barton-Gade, 1985), and genes of the major histocompatibility complex with many disease and productive traits in several species (see Lamont, 1989; Lewin, 1989; and Vaiman et al., 1988 for recent reviews in poultry, cattle, and pigs). Effects of transgenes on quantitative characters have also been measured (Pursel et al., 1987).

However, most of these quantitative traits are likely influenced also by many other genes that have smaller individual effects but large aggregate effects on the trait. Under certain population and breeding structures, these polygenic effects can be confounded with effects of the single gene or transgene under investigation, even in the absence of linkage. In the analysis of data to examine effects of single genes or transgenes on quantitative traits, it is important to disentangle this confounding to obtain unbiased estimates of single-gene effects and valid significant levels of tests of hypotheses about these single-gene effects. Most studies to date have used least squares analyses to measure single-gene effects and have ignored some or all of the effects of the background polygenes. Examples are the studies previously cited in which genetic variants of casein were associated with milk production traits (Ng-Kwai-Hang et al., 1984) and the gene for halothane

[1]Financial support was provided by the Nat. Sci. and Engineering Res. Council of Canada and the Canadian Assoc. of Anim. Breeders.

Received September 20, 1991.

Accepted February 10, 1992.

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reaction was associated with carcass traits in pigs (Jensen and Barton-Gade, 1985).

Kennedy and Van Arendonk (1989) and Kennedy and Schaeffer (1990) suggested that use of mixed-model methods under an individual animal model with inclusion of the identifiable single genotype as a fixed effect could separate effects of the single gene from those of the polygenes. In this paper we consider both randomly mated and selected populations and show where use of an individual animal model provides unbiased estimates and exact tests of hypotheses of single-gene effects and where use of ordinary least squares (OLS) analysis, the usual method of analysis, does not.

Methodology

Consider the following simple genetic model:

$$y_{ij} = g_i + a_{ij} + e_{ij} \qquad [1]$$

where y_{ij} is an observation on a quantitative trait on the ij^{th} animal, g_i is the effect of the ith observable genotype at a single locus, a_{ij} is the additive polygenic effect of the ij^{th} animal $\sim N(\mathbf{O}, \mathbf{A}\sigma_a^2)$, where **A** is a matrix of additive relationships among animals, and e_{ij} is a random environmental effect $\sim N(\mathbf{O}, \mathbf{I}\sigma_e^2)$. This is the classic mixed model both in a genetic sense of involving a mixture of polygenic and single-locus effects (e.g., Elston and Stewart, 1971; Morton and MacLean, 1974) and in a statistical sense of involving both fixed (single-locus genotype) and random (polygenic) effects (e.g., Searle, 1971). We are dealing with a special case of the genetic mixed model, however, in that we assume we can identify all individuals with observations on the quantitative trait as to genotype at the single locus of interest. For this model we assume also, for simplicity, no other fixed effects, polygenic effects are additive, and there are no interactions between single-gene and polygenic effects, but expansion of the model to accommodate more complicated models is possible and is discussed later.

In matrix notation, the model is as follows:

$$\mathbf{y} = \mathbf{Xg} + \mathbf{Za} + \mathbf{e} \qquad [2]$$

with (assuming no selection)

$$E(\mathbf{y}) = \mathbf{Xg}$$

and

$$\mathrm{Var}(\mathbf{y}) = \mathbf{V} = \mathbf{ZAZ}'\sigma_a^2 + \mathbf{I}\sigma_e^2. \qquad [3]$$

The matrix **X** assigns observations to single-locus genotypes (**g**) and the matrix **Z** assigns observations to animals. The matrix **Z** allows for animals without observations (e.g., ancestors) to be represented in **a** and **A**. If all animals in **a** have a single record, then **Z** = **I**, the identity matrix. Equations [2] and [3] represent an individual animal model (Henderson, 1984, 1988).

Mixed-Model Method

Estimation of effect of single-locus genotype under [2] and [3] are obtained from Henderson's (1973) mixed-model equations (Kennedy and Schaeffer, 1990) as follows:

$$\begin{bmatrix} \hat{\mathbf{g}} \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\mathbf{A}^{-1}\lambda \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X'y} \\ \mathbf{Z'y} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{X'y} \\ \mathbf{Z'y} \end{bmatrix} \qquad [4]$$

where

$$\lambda = \sigma_e^2/\sigma_a^2$$

or as

$$\hat{\mathbf{g}} = (\mathbf{X'V}^{-1}\mathbf{X})^{-1}\ \mathbf{X'V}^{-1}\mathbf{y}.$$

$\mathrm{Var}(\hat{\mathbf{g}})$ is computed as

$$\mathrm{Var}(\hat{\mathbf{g}}) = \mathbf{C}_{11}\sigma_e^2 \qquad [5]$$

and an estimate is provided by substituting

$$\hat{\sigma}_e^2 = (\mathbf{y'y} - \hat{\mathbf{g}}'\mathbf{X'y} - \hat{\mathbf{a}}'\mathbf{Z'y})/[N - r(\mathbf{X})]$$
$$= (\mathbf{y'V}^{-1}\mathbf{y} - \hat{\mathbf{g}}'\mathbf{X'V}^{-1}\mathbf{y})/[N - r(\mathbf{X})],$$

where N = the number of observations and $r(\mathbf{X})$ = the number of genotypes.

Henderson (1984) has indicated the null hypothesis H_o: $\mathbf{K'g} = 0$ can be tested from $F = Q/f\hat{\sigma}_e^2$, where $f = r(\mathbf{K})$ and

$$Q = (\mathbf{K}'\hat{\mathbf{g}})'(\mathbf{K'C}_{11}\mathbf{K})^{-1}(\mathbf{K}'\hat{\mathbf{g}}).$$

The *F*-test has f numerator and $N - r(\mathbf{X})$ denominator df.

Ordinary Least Squares

Frequently, in analysis of data for single gene effects an operational model of

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$$y_{ij} = g_i + \varepsilon_{ij},$$

where $\varepsilon_{ij} = a_{ij} + e_{ij}$ from [1] is assumed, and analysis is by OLS. In matrix notation, the model is as follows:

$$\mathbf{y} = \mathbf{Xg} + \boldsymbol{\epsilon}$$

with E(**y**) and Var (**y**) as in [2] and [3].

However, with OLS, Var(**y**) is (incorrectly) assumed for computation purposes to be $\mathbf{I}\sigma^2_\varepsilon$, where $\sigma^2_\varepsilon = \sigma^2_a + \sigma^2_e$ and the correlated error structure from $\mathbf{ZAZ}'\sigma^2_a$ is ignored. Under OLS, estimation of genotype effects is from

$$\hat{\mathbf{g}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \qquad [6]$$

Var($\hat{\mathbf{g}}$) is computed as follows:

$$\mathrm{Var}(\hat{\mathbf{g}}) = (\mathbf{X}'\mathbf{X})^{-1}\hat{\sigma}^2_\varepsilon,$$

where

$$\hat{\sigma}^2_\varepsilon = (\mathbf{y}'\mathbf{y} - \hat{\mathbf{g}}'\mathbf{X}'\mathbf{y})/[N - r(\mathbf{X})].$$

The hypothesis H_o: $\mathbf{K}'\mathbf{g} = \mathbf{0}$ is tested by $F = Q/f\hat{\sigma}^2_\varepsilon$, where

$$Q = (\mathbf{K}'\hat{\mathbf{g}})'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}'\hat{\mathbf{g}}),$$

with f numerator and N − r(**X**) denominator df.

Random Mating

We now consider properties of the mixed-model and OLS estimates under random mating.

Mixed Model

Under random mating, E(**y**) = **Xg**, and it is well known (e.g., Henderson, 1984) that (given λ)

$$E(\hat{\mathbf{g}}) = \mathbf{g}$$

and Var($\hat{\mathbf{g}}$) = $\mathbf{C}_{11}\sigma^2_e$ is minimum given E($\hat{\mathbf{g}}$) = **g** (i.e., $\hat{\mathbf{g}}$ is the best linear unbiased estimator [**BLUE**] of **g**). If inbreeding occurs, use of **A** in the mixed-model equations accounts for the decline in additive genetic variance associated with inbreeding (see Kennedy and Sorensen [1988] for a discussion).

Also, for

$$Q = \hat{\mathbf{g}}'\mathbf{C}_{11}^{-1}\hat{\mathbf{g}},$$

$$Q/f\hat{\sigma}^2_e \sim F[f, N - r(\mathbf{X})],$$

where f = the number of genotypes, gives an exact and best, in the sense of most powerful, test of the null hypothesis

$$H_o: \mathbf{g} = \mathbf{0}.$$

Linear functions of **g** can be tested using

$$Q = (\mathbf{K}'\hat{\mathbf{g}})'(\mathbf{K}'\mathbf{C}_{11}\mathbf{K})^{-1}(\mathbf{K}'\hat{\mathbf{g}})$$

where

$$f = r(\mathbf{K}).$$

Use of **A** in the estimation procedure accounts for the correlated structure among observations (**y**).

Ordinary Least Squares

With analysis by OLS,

$$E(\hat{\mathbf{g}}) = E(\mathbf{g}),$$

that is, estimates of single-gene effects are unbiased, but $\mathrm{Var}(\hat{\mathbf{g}}) \neq (\mathbf{X}'\mathbf{X})^{-1}\sigma^2_\varepsilon$, as would be computed by most OLS statistical software (Henderson, 1975a). Rather,

$$\begin{aligned}\mathrm{Var}(\hat{\mathbf{g}}) &= \mathrm{Var}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\left(\mathbf{I}\sigma^2_e + \mathbf{ZAZ}'\sigma^2_a\right)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\sigma^2_e + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{ZAZ}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\sigma^2_a.\end{aligned}$$

Only if **Z** = **I** and **A** = **I**, which is the case only if animals are unrelated and there is a single observation per animal, or if $\sigma^2_a = 0$, that is, there is no genetic variance due to polygenes, does $\mathrm{Var}(\hat{\mathbf{g}}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2_\varepsilon$. If these conditions are not true $(\mathbf{X}'\mathbf{X})^{-1}\sigma^2_\varepsilon$ will not represent true Var($\hat{\mathbf{g}}$).

This also has ramifications for hypothesis tests involving **g**. For

$$Q = (\mathbf{K}'\hat{\mathbf{g}})'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}'\hat{\mathbf{g}})$$

and

$$\hat{\sigma}^2_\varepsilon = (\mathbf{y}'\mathbf{y} - \hat{\mathbf{g}}'\mathbf{X}'\mathbf{y})/[N - r(\mathbf{X})],$$

$$Q/f\hat{\sigma}^2_\varepsilon \nsim F[f, N - r(\mathbf{X})]$$

under the null hypothesis H_o: $\mathbf{K}'\mathbf{g} = \mathbf{0}$, because Q and $f\hat{\sigma}^2_\varepsilon$ are not independent and do not have the same expectation. Indeed, if the true model is [2] and [3] it is likely that

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$$E(Q/f) > E\left(\hat{\sigma}_\varepsilon^2\right).$$

It can easily be shown that

$$\begin{aligned} E(\mathbf{y'y} - \hat{\mathbf{g}}'\mathbf{X'y}) &= E[\mathbf{y'}(\mathbf{I}-\mathbf{X}(\mathbf{X'X})^{-1}\mathbf{X'})\mathbf{y}] \\ &= tr[(\mathbf{I}-\mathbf{X}(\mathbf{X'X})^{-1}\mathbf{X'})\left(\mathbf{I}\sigma_e^2 + \mathbf{ZAZ'}\sigma_a^2\right)] \\ &= [N - r(\mathbf{X})]\sigma_e^2 + tr[\mathbf{ZAZ'} - \mathbf{X}(\mathbf{X'X})^{-1}\mathbf{X'ZAZ'}]\sigma_a^2. \end{aligned}$$

Now

$$tr(\mathbf{ZAZ'}) = tr(\mathbf{AZ'Z}) = \sum_{k=1}^{n} n_k a_k,$$

where a_k is the kth diagonal element of **A** and n_k is the number of records on individual k. In the absence of inbreeding $\Sigma n_k a_k = N$, and with inbreeding $\Sigma n_k a_k = N + \Sigma n_k F_k$, which reduces to $N + \Sigma F_k$ when each individual has a single record, where F_k is the inbreeding coefficient of k. Now the second term involving σ_a^2, $tr[\mathbf{X}(\mathbf{X'X})^{-1}\mathbf{X'ZAZ'}] = tr[\mathbf{AZ'X}(\mathbf{X'X})^{-1}\mathbf{X'Z}]$. If **X** and **Z** are ordered by genotype, and when each individual has a single record, $\mathbf{Z'X}(\mathbf{X'X})^{-1}\mathbf{X'Z} = \bigoplus_{i=1}^{t} \mathbf{J}_i/n_{i.}$, where t is the number of genotypes, $\oplus$ represents a direct sum, $\mathbf{J}_i$ is a matrix with each element equal to one and $n_{i.}$ is the number of individuals with genotype i. Therefore,

$$\begin{aligned} tr\left(\mathbf{A} \bigoplus_{i=1}^{t} \mathbf{J}_i/n_{i.}\right) &= \sum_{i=1}^{t} n^{-1}_{i.} \sum_{j=1}^{n_{i.}} \sum_{k=1}^{n_{i.}} a_{ijk} \\ &= \sum_{i=1}^{t} n^{-1}_{i.} n^2_{i.}\, \bar{a}_{ii} \\ &= \sum_{i=1}^{t} n_{i.}\bar{a}_{ii} \end{aligned}$$

where $\bar{a}_{ii}$ is the average relationship among individuals of genotype i. (Results are similar if we relax the assumption that each individual has a single record, except that individuals contribute to $\bar{a}_{ii}$ in proportion to their number of records, that is, $\bar{a}_{ii}$ is an average relationship weighted by number of records contributed by each animal.)

Therefore, the expectation of σ_ε^2, again assuming for simplicity that each individual has a single record, is

$$\begin{aligned} &([N - r(\mathbf{X})]\sigma_e^2 + [N + \sum_{k=1}^{n} F_k - \sum_{i=1}^{t} n_{i.}\bar{a}_{ii}]\sigma_a^2)/[N \\ &- r(\mathbf{X})] = \sigma_e^2 + (N + \Sigma F_k - \Sigma n_{i.}\bar{a}_{ii})\sigma_a^2/[N - r(\mathbf{X})]. \end{aligned}$$

Note that if all individuals are unrelated and there is no inbreeding $(N + \Sigma F_k - \Sigma n_{i.}\bar{a}_{ii}) = N - t = N - r(\mathbf{X})$ and $E\left(\hat{\sigma}_\varepsilon^2\right) = \sigma_e^2 + \sigma_a^2$ (i.e., $\hat{\sigma}_\varepsilon^2$ is an unbiased estimator of σ_ε^2). However, if animals are related within genotype, $\Sigma n_{i.}\bar{a}_{ii} > t$ and there is a downward bias in $\hat{\sigma}_\varepsilon^2$, assuming no inbreeding. With inbreeding the direction of the bias in $\hat{\sigma}_\varepsilon^2$ depends on whether $\Sigma F_i - \Sigma n_{i.}\bar{a}_{ii}$ is > or < −t. A downward bias in $\hat{\sigma}_\varepsilon^2$, unless accompanied by an equal proportional downward bias in the contribution of σ_ε^2 to Q, will result in an inflated *F*-test. This is consistent with the caution given by statisticians that "positive correlations among the errors within a treatment vitiate the *F*-test, giving too many significant results" (Snedecor and Cochran, 1967, p 323).

Now we examine Q, the numerator of the *F*-test. Under H_o: $\mathbf{K'g} = \mathbf{0}$, $E(Q) = tr([\mathbf{K'}(\mathbf{X'X})^{-1}\mathbf{K}]^{-1}\mathbf{K'}[\mathbf{X'X}]^{-1}\mathbf{X'ZAZ'X}[\mathbf{X'X}]^{-1}\mathbf{K})\sigma_a^2 + tr(\mathbf{I})\sigma_e^2$. Again, if individuals are ordered by genotype and each individual has a single observation, it can be shown that $(\mathbf{X'X})^{-1}\mathbf{X'ZAZ'X}(\mathbf{X'X})^{-1} = \bar{\mathbf{A}} = \{\bar{a}_{ij}\}$ for i,j = 1, 2, ... t. Consider now some hypotheses associated with estimates of genotype effects of interest. We might have two alleles and three genotypes and be interested, for example, in the estimate of the difference between the two homozygous genotypes and the test of the hypothesis that the genotypes are equal. In this instance $\mathbf{K'} = (1\ 0\ -1)$ and $E(Q) = \sigma_e^2 + (1/n_{1.} + 1/n_{3.})^{-1}(\bar{a}_{11} + \bar{a}_{33} - 2\bar{a}_{13})\sigma_a^2$. If individuals are unrelated $\bar{a}_{11} = 1/n_{1.}$, $\bar{a}_{33} = 1/n_{3.}$, and $\bar{a}_{13} = 0$, and therefore $E(Q) = \sigma_e^2 + \sigma_a^2 = \sigma_\varepsilon^2$ and the *F*-test is unbiased. However, if individuals are related there is a bias. If individuals within a genotype are more highly related than individuals of different genotypes, as is likely the case, then the bias is upward, that is, $(\bar{a}_{11} + \bar{a}_{33} - 2\bar{a}_{13}) > (1/n_{1.} + 1/n_{3.})$. Recall that the denominator of the *F*-test, $\hat{\sigma}_\varepsilon^2$, is biased downward in the presence of relationships within genotype (ignoring inbreeding), and, therefore, it is likely that the *F*-test will be inflated and give too many "significant" results. The estimate of the difference between genotypes, in the absence of selection, is nonetheless unbiased.

Similarly, the test that the heterozygous genotype is equal to the average of the homozygous

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genotypes (i.e., dominance = 0) can be constructed through $\mathbf{K}' = (-1\ 2\ -1)$. With this hypothesis, it can be shown under the null hypothesis that $E(Q) = \sigma_e^2 + (1/n_{1.} + 4/n_{2.} + 1/n_{3.})^{-1}(\bar{a}_{11} + 4\bar{a}_{22} + \bar{a}_{33} - 4\bar{a}_{12} + 2\bar{a}_{13} - 4\bar{a}_{23})\sigma_a^2$. Again, if individuals are unrelated, $E(Q) = \sigma_e^2 + \sigma_a^2 = \sigma_\varepsilon^2$, but if they are related and relationships are greater within genotype than across genotype, then there is likely to be an upward bias in Q and again a likelihood of too many significant results. Similarly, a test of the hypothesis $g_1 = g_2 = g_3$, which can be constructed from the two previous tests, will likely be biased upward. These potential biases are illustrated later through computer simulation.

Selection

Again for simplicity, assume that the single locus has two allelic forms and three possible genotypes. Assume also that at Generation 0 the population is in Hardy-Weinberg and linkage equilibrium and let genotypes g_1, g_2, and g_3 have values of α, δ, and $-\alpha$ and frequencies of p^2_0, $2p_0q_0$, and q^2_0, respectively. If $\mathbf{y}_0$ represents phenotypic observations then, before selection, $E(\mathbf{y}_0) = \mathbf{Xg}$. We now impose truncation selection on $\mathbf{y}_0$ such that individuals with the highest phenotypic values are selected to produce progeny in Generation 1, and those with the lowest phenotypic values do not contribute progeny.

For the jth selected individual of genotype i,

$$E\left(y_{ij_s}\right) = g_i + v_i + w_i,$$

where $v_i = h^2E\left(\bar{y}_{i.s} - \bar{y}_{i.0}\right)$ and $w_i = (1 - h^2)E\left(\bar{y}_{i.s} - \bar{y}_{i.0}\right)$ are the expected additive genetic and environmental values of selected individuals of the ith genotype. Because new environmental deviations are generated each generation, w_i does not concern us for Generation 1, but v_i does. For $\alpha \geq \delta \geq -\alpha$, then $v_1 \leq v_2 \leq v_3$. Intuitively this seems reasonable because on average, if an individual of the poorest single genotype is selected, it is likely to be better than average for additive genetic value of polygenes, and vice versa. In matrix notation, the expected value of the vector of observations above the truncation point is

$$E\left(\mathbf{y}_{0_s}\right) = \mathbf{X}_s\mathbf{g} + \mathbf{v} + \mathbf{w}$$

for $\mathbf{v}' = (v_1\ v_2\ v_3)$ and $\mathbf{w}' = (w_1\ w_2\ w_3)$. The values of $\mathbf{X}_s$, $\mathbf{v}$, and $\mathbf{w}$ will depend on initial gene frequencies, the proportion of individuals selected, and values of $\mathbf{g}$, σ_a, and σ_e. If the proportional contributions of genotypes g_1, g_2, and g_3 from Generation 0 to 1 are p^2_0r, $2p_0q_0s$, and q^2_0t, respectively, where r, s, and t are the relative fitnesses of the respective genotypes scaled to keep population size constant, then in Generation 1 the expected value of an individual of the ith genotype, if matings among selected parents is at random, is

$$E(y_{11}) = \alpha + p_0(p_0rv_1 + q_0sv_2)/p_1 \quad [7]$$

$$E(y_{21}) = \delta + (p^2_0q_1rv_1 + p_0q_0sv_2 + q^2_0p_1tv_3)/2p_1q_1 \quad [8]$$

$$E(y_{31}) = -\alpha + q_0(p_0sv_2 + q_0tv_3)/q_1 \quad [9]$$

with

$$p_1 = p^2_0r + p_0q_0s$$

and

$$q_1 = p_0q_0s + q^2_0t.$$

Truncation selection also reduces the genetic variance of the polygenes in Generation 1 due to gametic phase or linkage disequilibrium (Bulmer, 1971).

We now consider the effects of this selection on estimates of $\mathbf{g}$ from mixed-model methods and OLS.

Mixed-Model Methods

Properties of mixed-model methods when selection has occurred are not fully understood. Use of $\mathbf{A}$ in [4] accommodates the change in variance due to gametic disequilibrium (Kennedy and Sorensen, 1988). Henderson (1975b) has shown that use of the usual mixed-model equations [4], ignoring selection, yields BLUE of fixed effects and BLUP of random effects if selection is on a linear function of the data, which he designated as $\mathbf{L'y}$, $\mathbf{a}$ and $\mathbf{e}$ are multivariate normal, which implies an infinitesimal model for the polygenes, $\mathbf{A}$ and λ are known, and $\mathbf{L'X} = \mathbf{0}$. It is this last restriction that is most troublesome. Henderson (1973) has interpreted this to imply that selection must be within levels of fixed effects.

If we assume the effect of the single-locus genotype to be fixed, the selection process envisioned here clearly is not within genotype at the single-genetic locus and seemingly does not fit Henderson's requirement that $\mathbf{L'X} = \mathbf{0}$. Gianola and Fernando (1986) have argued, however, that from a Bayesian viewpoint this requirement does not arise. Thompson (1979) seems to question Henderson's interpretation that $\mathbf{L'X} = \mathbf{0}$ implies selection must be within fixed effects and provides

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a counter example of selection ignoring fixed effects. He argues further that, with repeated sampling, **L** is not fixed and will vary from sample to sample. In our sampling scheme, so will **X**. Additional discussion on these points is in the papers of Gianola et al. (1989), Im et al. (1989), and Kennedy (1991).

It seems intuitively reasonable that, if the data used for analyses contain all the observations used in the selection decision process, use of [4] will provide unbiased estimates of **g**, over repeated sampling, even though selection is directly on the observations (**y**) that include the contributions of the effects of genotype at the single-genetic locus. However, if data on which selection was practiced are not included, some bias might result. We subsequently test this through computer simulation to verify that use of [4] provides unbiased estimates and correct sampling variances of $\hat{\mathbf{g}}$ and exact tests of H_0: $\mathbf{K}'\hat{\mathbf{g}} = \mathbf{0}$ when all data are used in the analysis.

Ordinary Least Squares

We now consider properties of OLS estimates of **g** under selection. First consider the situation in which only data from the progeny of selected parents are used (Generation 1). From [7] and [9] it is easy to show that the estimate of α as obtained from $\hat{\alpha} = 1/2(\bar{y}_{11} - \bar{y}_{31})$, which is the estimate that would be obtained by OLS, is biased by $1/2[p_0(p_0rv_1 + q_0sv_2)/p_1 - q_0(p_0sv_2 + q_0tv_3)/q_1]$. Because with directional selection $p_1 > p_0$ and $q_1 < q_0$ and $v_1 < v_2 < v_3$, this bias is negative, that is, the effect of the single gene is underestimated.

These biases are illustrated in Figure 1 for additive (Figure 1a), dominance (Figure 1b), and recessive (Figure 1c) models. Biases are given for a range of selection intensities (1, 10, 50, 90%) and initial frequencies of the favorable gene (.05 to .95) for a single gene with a large true effect (α = .5 SD). Dominance effects were either complete (δ = .5), none (δ = 0), or recessive (δ = −.5). In all cases, the bias is directly proportional to heritability of the polygenes. Therefore, the biases given in Figure 1 are for h^2 = 1 and values for any other heritability can be obtained simply by multiplying the value in the figure by h^2. Under an additive model (δ = 0) the bias depends mostly on selection intensity and increases with increasing intensity. With complete dominance, ($\delta = \alpha$), the bias is greatest for small initial frequencies of the favorable gene and tends to zero at high gene frequencies for all intensities of selection. With a recessive model ($\delta = -\alpha$), the opposite tends to occur.

If δ is estimated as $\hat{\delta} = \bar{y}_{21} - 1/2(\bar{y}_{11} + \bar{y}_{31})$, it can be easily shown from [7], [8], and [9] that $E(\hat{\delta}) = \delta$ and there is no bias.

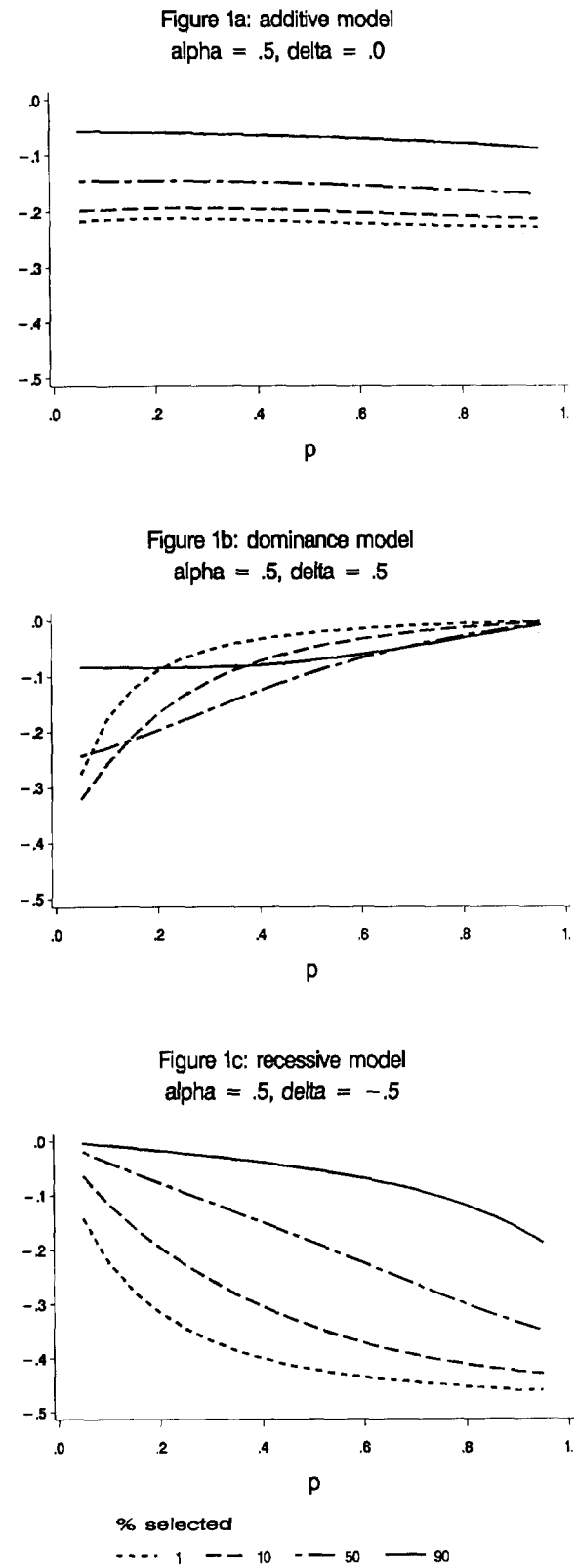

Figure 1. Bias in estimates of α by least squares after one generation of selection under additive, dominance, and recessive models for genes of large (α = .5 SD) effects using data from the progeny generation only. Proportion selected is 1, 10, 50, or 90% and gene frequency ranges from p = .05 to p = .95. (Results are expressed for a trait with phenotypic SD = 1 and h^2 = 1.)

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If data are pooled over Generations 0 and 1, and all data of Generation 0 are included, the situation is more complicated. The OLS estimate of g_1 is

$$\hat{g}_1 = (p^2_0\bar{y}_{10} + p^2_1\bar{y}_{11})/(p^2_0 + p^2_1)$$

and

$$E(\hat{g}_1) = (p^2_0\alpha + p^2_1[\alpha + p_0(p_0rv_1 + q_0sv_2)/p_1])/(p^2_0 + p^2_1) = \alpha + p_0p_1(p_0rv_1 + q_0sv_2)/(p^2_0 + p^2_1).$$

The latter term represents the bias in $\hat{g}_1$. This bias is in the direction of selection, that is, if positive selection is practiced, v_1 and v_2 will be positive.

Similarly, the expected value of the OLS estimate of g_3 is

$$E(\hat{g}_3) = -\alpha + q_0q_1(p_0sv_2 + q_0tv_3)/(q^2_0 + q^2_1).$$

Therefore, if we estimate α from OLS estimates of $1/2(g_1 - g_3)$, that is, $\mathbf{K'g}$ for $\mathbf{K'} = (1/2\ 0 - 1/2)$, then

$$E(\hat{\alpha}) = \alpha + p_0p_1(p_0rv_1 + q_0sv_2)/2(p^2_0 + p^2_1) - q_0q_1(p_0sv_2 + q_0tv_3)/2(q^2_0 + q^2_1). \quad [10]$$

As selection progresses, p will increase relative to q and will tend to cause a positive bias in $\hat{\alpha}$. The numerator terms in the bias can be expressed as $p_1(z_1 + 1/2\ z_2)$ and $q_1(1/2\ z_2 + z_3)$, where $z_1 = p^2_0rv_1$, $z_2 = 2p_0q_0sv_2$, and $z_3 = q^2_0tv_3$. These terms (z_i) are h^2 times the genotypic frequency times the probability density of each genotype at its selection truncation point (divided by the proportion of all individuals in Generation 0 selected to be parents). The bias in $\hat{\alpha}$ is then $p_1(z_1 + 1/2\ z_2)/2(p^2_0 + p^2_1) - q_1(1/2\ z_2 + z_3)/2(q^2_0 + q^2_1)$.

The estimate of δ requires an estimate of g_2. The OLS estimate of g_2 is

$$\hat{g}_2 = (p_0q_0\bar{y}_{20} + p_1q_1\bar{y}_{21})/(p_0q_0 + p_1q_1)$$

and

$$E(\hat{g}_2) = \delta + (p^2_0q_1rv_1 + p_0q_0sv_2 + q^2_0p_1tv_3)/2(p_0q_0 + p_1q_1).$$

If we estimate δ from OLS estimates of $g_2 - 1/2(g_1 + g_3)$, that is, $\mathbf{K'g}$ for $\mathbf{K'} = (-1/2\ 1\ -1/2)$, then

$$E(\hat{\delta}) = \delta + (p^2_0q_1rv_1 + p_0q_0sv_2 + q^2_0p_1tv_3)/2(p_0q_0 + p_1q_1) - p_0p_1(p_0rv_1 + q_0sv_2)/2(p^2_0 + p^2_1) - q_0q_1(p_0sv_2 + q_0tv_3)/2(q^2_0 + q^2_1) = (q_1z_1 + 1/2\ z_2 + p_1z_3)/2(p_0q_0 + p_1q_1) - p_1(z_1 + 1/2\ z_2)/2(p^2_0 + p^2_1) - q_1(1/2\ z_2 + z_3)/2(q^2_0 + q^2_1). \quad [11]$$

Biases in $\hat{\alpha}$ and $\hat{\delta}$ are pictured in Figures 2a to 2f and 3a to 3f for a range of selection intensities (1, 10, 50, 90%), initial frequencies of the favorable gene (.05 to .95), and true effects of α and δ. The true effects of single genes examined were large (α = .5 SD) and moderate (α = .25 SD). Dominance effects were either complete (δ = .5 or .25, depending on α), none (δ = 0), or recessive (δ = −.5 or −.25).

The biases are expressed relative to a heritability of 1. Again, because the bias is directly proportional to heritability, biases for heritabilities < 1 can be obtained by simply multiplying the bias depicted in the figure by h^2.

Bias in $\hat{\alpha}$ is influenced greatly by selection and is generally positive with intense selection (1 and 10%). At low selection intensities (50 and 90%), bias in $\hat{\alpha}$ is small. Under an additive model at very high selection intensity (1%), bias (assuming h^2 = 1) is usually greater than the size of the true effect (Figures 2a and 2d) but decreases proportionally as the magnitude of the true effect increases. Bias is slightly higher at high gene frequency than at low. With dominance and high intensities of selection, bias is more dependent on gene frequency, peaks at frequencies of the desirable gene of between .1 and .2, and then drops off quickly as gene frequency increases (Figures 2b and 2e). Again, bias can be very large and greater than the true effect, particularly at gene frequencies of less than .5. With recessive gene action, bias tends to increase with gene frequency (Figures 2c and 2f). At high intensity of selection and high gene frequency, bias can be about twice the size of the true effect (for h^2 = 1).

As with $\hat{\alpha}$, bias in $\hat{\delta}$ (Figures 3a to 3f) is small with weak selection (50 and 90%). With intense selection, direction of the bias depends very much on mode of gene action and gene frequency. Under an additive model (Figures 3a and 3d), large positive biases in $\hat{\delta}$ occur at low gene frequency and large negative biases occur at high gene frequency if selection is very intense (1%). Otherwise, bias in $\hat{\delta}$ is relatively small. With complete dominance (Figures 3b and 3e), bias in $\hat{\delta}$ is largely positive and can be high at low gene frequency if selection intensity is high, but it tends to zero at intermediate and high gene frequencies. However, with a recessive model (Figures 3c and 3f), large negative biases in $\hat{\delta}$ can occur at high gene frequency and intense selection, with small positive biases at low gene frequencies. In contrast to the situation in estimation of α, bias in $\hat{\delta}$ tends to be proportionally less for genes of moderate effect than for genes of large effect.

The preceding is based on the assumption that there is an effect of the single gene (i.e., $\alpha \neq 0$). If $\alpha = \delta = 0$, then $v_1 = v_2 = v_3$, $r = s = t$, $p_1 = p_0$, and

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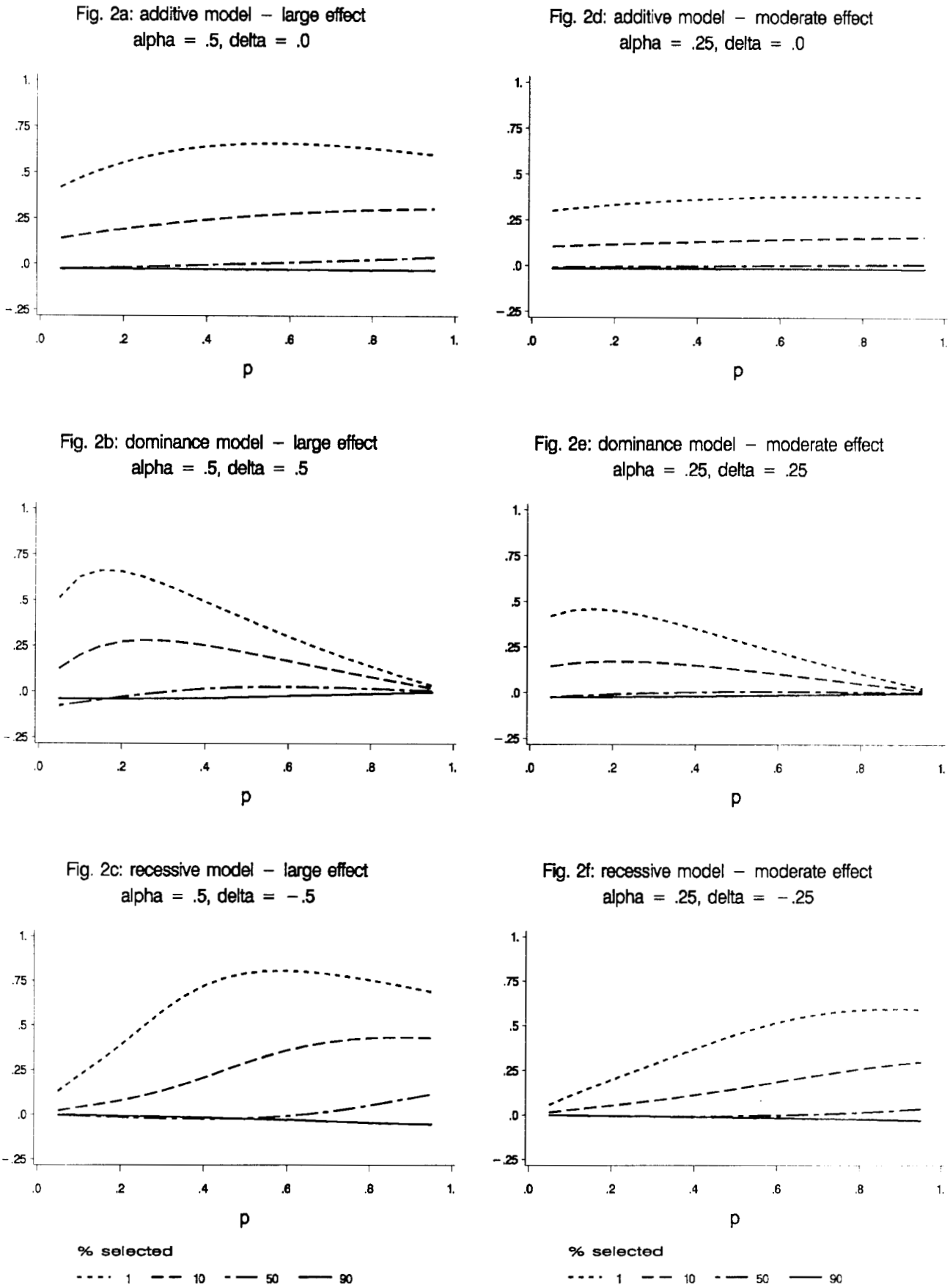

Figure 2. Bias in estimates of α by least squares after one generation of selection under additive, dominance, and recessive models for genes of large (α = .5 SD) and moderate (α = .25 SD) effects using data from both the parental and progeny generations. Proportion selected is 1, 10, 50, or 90% and gene frequency ranges from p = .05 to p = .95. (Results are expressed for a trait with phenotypic SD = 1 and h^2 = 1.)

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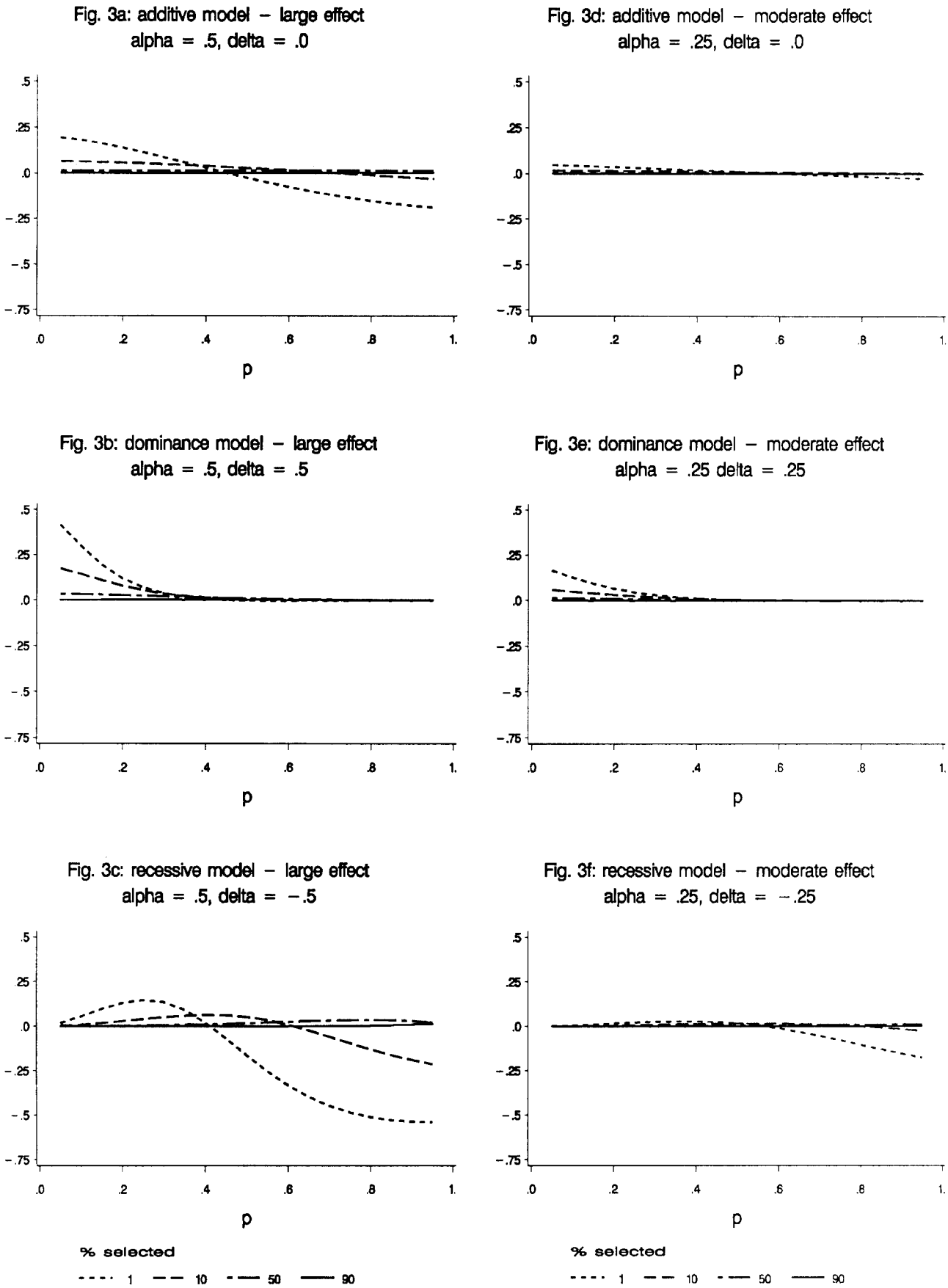

Figure 3. Bias in estimates of δ by least squares after one generation of selection under additive, dominance, and recessive models for genes of large (α = .5 SD) and moderate (α = .25 SD) effects using data from both the parental and progeny generations. Proportion selected is 1, 10, 50, or 90% and gene frequency ranges from p = .05 to p = .95. (Results are expressed for a trait with phenotypic SD = 1 and h^2 = 1.)

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Table 1. Frequencies of rejecting hypotheses $\alpha = 0$ and $\delta = 0$ using mixed-model (MM) and ordinary least squares (OLS) procedures with up to five generations of random and phenotypic selection when true $\alpha = 0$ and $\delta = 0$

	Analysis across generations				Analysis within generations			
	H_0:α = 0		H_0:δ = 0		H_0:α = 0		H_0:δ = 0	
Generation	MM	OLS	MM	OLS	MM	OLS	MM	OLS
Random selection[a]								
0	.030[c]	.030	.060	.060	.030	.030	.060	.060
1	.050	.065	.040	.045	.076	.086*	.061	.056
2	.040	.105*	.060	.055	.041	.097*	.062	.072
3	.030	.140**	.060	.055	.026	.046	.057	.067
4	.045	.130**	.055	.070	.053	.079	.053	.074
5	.035	.150**	.045	.075	.026	.048	.048	.053
Phenotypic selection[b]								
0	.030	.030	.060	.060	.030	.030	.060	.060
1	.050	.070	.040	.050	.045	.060	.065	.065
2	.050	.100*	.055	.050	.046	.071	.051	.081
3	.040	.140**	.035	.050	.071	.096*	.030	.046
4	.040	.225**	.055	.045	.063	.109*	.052	.063
5	.035	.290**	.050	.045	.022	.044	.028	.038

[a]Random selection of 5 of 20 males, no selection of females (h^2 = .5).
[b]Selection of best 5 of 20 males on phenotype, no selection of females (h^2 = .5).
[c]Frequency of rejecting H_0 with assumed Type I error rate of .05 over 200 replicates.
*P < .05.
**P < .01.

$q_1 = q_0$. If we let $v^* = rv_1 = sv_2 = tv_3$, then it is simple to show that [7], [8], and [9], respectively, reduce to $E(y_{11}) = E(y_{21}) = E(y_{31}) = v^*$, that is, the expected value of an observation on each genotype at Generation 1 is simply the response to selection for the polygenes. As a result, $E(\hat{\alpha})$ of [10] reduces to α and $E(\hat{\delta})$ of [11] reduces to δ, that is, there is no bias in the least squares estimates of α and δ whether data from both generations are used or only data from Generation 1 are used. Although $\hat{\alpha}$ and $\hat{\delta}$ are unbiased when true $\alpha = \delta = 0$, we illustrate later through computer simulation that the likelihood of finding spurious "significant" effects of α and δ increases with selection.

Simulation

Estimation of effects of single genes and associated tests of hypothesis were investigated further over several generations of random mating or selection through computer simulation. A general-purpose genetic simulation program previously described by Quinton et al. (1992) was used.

The trait of interest had an initial mean of 0 and phenotypic standard deviation of 1. Heritability was .5. Initial gene frequency was .5. Population size was 40, with 20 males and 20 females. A single phenotypic record was simulated for each individual in the population. Each generation, five males were chosen either at random or on the basis of phenotype for the trait of interest and were mated at random to the 20 females to produce the next generation (two offspring per female). The simulation was conducted over five generations. For each selection scheme, 200 replicates were performed. Data were analyzed at the end of each generation by mixed-model methods [4] or by OLS [6] using all data pooled over all generations and using data of the most recent generation only.

Estimates of α and δ were measured for each selection scheme, and frequency of rejection of H_0: **Kg = 0** with a Type I error rate of .05 was measured for situations in which there was no true effect of the gene. Empirical standard errors of estimates of $\hat{\alpha}$ and $\hat{\delta}$ were used to test for significant bias by t-test and excess frequency of rejection (> .05) of tests of hypotheses was tested by χ^2.

In the first situation considered, there was no effect of the single gene ($\alpha = 0$, $\delta = 0$). As expected, with both random and phenotypic selection mixed-model procedures and OLS both gave unbiased estimates of α and δ. Use of mixed-model procedures gave frequencies of rejection of H_0: $\alpha = 0$ and H_0: $\delta = 0$ close to expected frequencies of .05. However, from Generation 2 on, when use was made of all the data, OLS analysis rejected the hypothesis that $\alpha = 0$ more frequently than the expected frequency of .05 for the Type I error rate specified (Table 1). By Generation 5 the empirical Type I error rate was .15. With phenotypic

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selection, frequency of rejection of the null hypothesis $\alpha = 0$ increased further to .29 by Generation 5. When analysis was performed within generation, however, frequency of rejection $\alpha = 0$ was close to .05. There was a marginally significant excess of frequency of rejection with the within-generation analysis for only a few of the generations. Frequency of rejection of H_0: $\delta = 0$ was close to the expected value of .05 in all analyses by OLS.

In the second situation considered, there was a real effect of the single gene ($\alpha = \delta = .5$). That is, a dominance model was assumed and the gene had a major effect. With random selection, again there was no bias in $\hat{\alpha}$ and $\hat{\delta}$ with both mixed-model and OLS procedures. However, with selection, the effects of the gene were overestimated with OLS but not with mixed models when data were pooled over generations (Table 2). By Generation 3, the overestimate in $\hat{\alpha}$ with OLS was 6% and by Generation 5 it was 19%. The bias would be expected to be greater at lower gene frequencies and higher intensities of selection (Figures 1b and 1e). Bias in $\hat{\delta}$ was not detected. Expected bias in $\hat{\delta}$ would be small at this gene frequency (Figures 3b and 3e). Analysis within generation showed no detectable bias in $\hat{\alpha}$ with OLS. A negative bias would be expected, but at a gene frequency of .5 the bias is expected to be fairly small and would be difficult to detect.

Discussion

Most studies of effects of single genes on quantitative traits have used OLS methods for estimation and hypothesis testing. Such analyses usually ignore all or some of the relationships among animals due to polygenes (A). When relationships involving polygenic effects are ignored, we have shown that in the situation in which the gene has no real effect, the likelihood of finding spurious "significant" effects can be much higher than the assumed Type I error rate with least squares analysis, particularly if data involving several generations are used. This is true for both randomly selected populations and populations for which selection has been practiced for the trait of concern. In the latter case, frequency of spurious significant results can be very high, as illustrated through the simulation example where, when five generations of data were involved, the hypothesis that the gene had no effect was rejected (incorrectly) 29% of the time (Table 1). The problem is less severe when least squares analysis is within generation. It was shown that the numerator of the *F*-test is biased upward when relationships within genotype are greater than relationships between genotypes. This occurrence increases when data are pooled over generations and selection has occurred. In many studies using field or experimental data in livestock, data are likely to span several generations because of the overlapping generation breeding structures and data are frequently collected over time. This likely explains why it is not uncommon to see reports of finding many more significant associations of single gene effects than expected by chance in different populations, yet comparisons of results across populations show inconsistencies and contradictions (e.g., Jensen et al., 1968; Kennedy et al., 1973). If the gene truly has no effect, estimates of effects of the gene are, nonetheless, unbiased when OLS procedures are used.

If the gene truly has an effect, least squares estimates are still unbiased with random selection but not with directional selection for the trait of interest. We have shown that if only data on progeny of selected parents are used, the additive effect of the gene, but not the dominance effect, is underestimated with least squares analysis because selection response for polygenes is less for the better single genotypes. The bias is directly proportional to heritability of the polygenes. With an underlying additive model the bias depends mostly also on selection intensity and the effect of gene frequency is small. With complete dominance or recessive models, however, the bias depends more on gene frequency. As mentioned, the bias is always negative regardless of mode of gene action.

This is not so when data of both the selected parental and resulting progeny generations are used. Here the bias is largely, but not universally, positive for estimates of additive effects. This is the result of concurrent selection response for the polygenes and changes in gene frequency at the single gene locus. Again, biases are directly proportional to heritability of the polygenes. The bias is greatly influenced by selection intensity and by true size of effect in absolute terms. However, in relative terms, bias is larger for genes of smaller effect than for genes of larger effect. Gene frequency has a larger effect on bias of estimates of the additive effect under nonadditive genetic models than under additive genetic models. Bias in least squares estimates of dominance effects tends to be smaller than additive effect bias and tends to be small unless selection is very intense. The bias can be highly dependent on gene frequency and mode of gene action. However, except in some extreme situations, biases in estimates of dominance effects by OLS analysis are likely to be small.

As previously stated, most published estimates of single-gene effects were obtained by least squares. If selection, either artificial or natural,

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Table 2. Estimates of α and δ using mixed-model (MM) and ordinary least squares (OLS) procedures with up to five generations of phenotypic selection[a] when true α = .5 and δ = .5

	Analysis across generations				Analysis within generations			
	$\hat{\alpha}$		$\hat{\delta}$		$\hat{\alpha}$		$\hat{\delta}$	
Generation	MM	OLS	MM	OLS	MM	OLS	MM	OLS
0	.468	.468	.513	.513	.468	.468	.513	.513
	(.016)[b]	(.016)	(.021)	(.021)	(.016)	(.016)	(.021)	(.021)
1	.474	.479	.496	.505	.479	.467	.479	.474
	(.012)	(.013)	(.013)	(.014)	(.017)	(.018)	(.022)	(.023)
2	.482	.498	.506	.514	.474	.463	.513	.512
	(.011)	(.012)	(.011)	(.012)	(.019)	(.012)	(.023)	(.024)
3	.489	.531*	.497	.502	.506	.504	.478	.471
	(.010)	(.012)	(.010)	(.011)	(.023)	(.024)	(.027)	(.028)
4	.490	.560**	.506	.507	.477	.478	.533	.544
	(.009)	(.012)	(.010)	(.011)	(.020)	(.020)	(.027)	(.028)
5	.493	.593**	.509	.512	.502	.492	.538	.539
	(.009)	(.013)	(.009)	(.011)	(.027)	(.026)	(.031)	(.032)

[a]Selection of best 5 of 20 males on phenotype, no selection of females (h^2 = .5).
[b]Standard error of 200 replicates.
*P < .05.
**P < .01.

was practiced for the traits of concern, then the estimates are potentially biased and care should be exercised in interpreting them. This requires assessment of the amount of selection practiced, structure of the data, mode of inheritance, polygenic heritability, gene frequency, and actual size of the effect of the gene. The statistical model used in the least squares analysis is also a factor because some of the problems arising as a result of relationships among animals are accommodated when the least squares analysis includes effects due to sires and dams treated as fixed effects (Mallard et al., 1991).

Difficulties associated with use of least squares and those of spurious significant results when gene effects are nil can be avoided or mitigated through use of mixed-model techniques. We have argued that the mixed-model procedures give unbiased estimates of the single-gene effects and exact tests of associated hypotheses for both randomly and directionally selected populations if all data are used and heritability of the polygenes (at the start of selection) is known. Also, sampling variances of estimates are smallest. Computer simulation supported this. If heritability is not known, errors in heritability value assumed can change significance levels (Kennedy, 1991) and possibly lead to some bias in estimates of effects. However, computer simulation has indicated that when heritability of the polygenes is not known, unbiased estimates of parameters over repeated sampling will still result if they are estimated simultaneously with the single-gene effects through mixed-model analyses (Sorensen and Kennedy, 1986). If only some of the data are used, as for example with analysis of one generation of data, theoretically there are potential biases with mixed-model analyses if selection has been practiced, but these are likely to be small if all relationships are used in the analysis. Significant bias was not detected in the simulation study when mixed-model analyses within generation were used. The mixed-model analysis presupposes the availability of pedigree information on animals in the population, but this is common in both experimental and field populations of livestock.

Finally, this study has invoked, for simplicity of illustration, a fairly simple genetic model [1] with respect to the polygenes and of fixed effects. Real situations are likely to involve more complex models and data structures. Simplicity was assumed to make the algebraic results tractable. Nonetheless, the results are more general and the model can be readily expanded to accommodate many more complicated situations (Henderson, 1984; Kennedy and Schaeffer, 1990).

Implications

Many reports in the literature of "significant" effects of single genes on quantitative traits based on ordinary least squares analyses are likely spurious. Also, published reports of estimates of single-gene effects may be biased if selection has been practiced for the trait of concern, as would frequently be the case when estimates are obtained from field populations. The use of mixed-model procedures, which account for all known genetic relationships among animals, is recommended to mitigate these problems.

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